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Sang Hyeok Im (임상혁)

Curriculum Vitae

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EDUCATION

Ph.D student in Astronomy Program (2021.02 - present)

Department of Physics and Astronomy, Seoul National University, 1, Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea

B.S. in Physics (2015.03 - 2021.02; double major)

Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, Republic of Korea

B.S. in Astronomy and Atmospheric Sciences (2015.03 - 2021.02; double major)

Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, Republic of Korea

PUBLICATIONS

Refereed Journal Article

First Author

1. Testing Ly α Emitters and Lyman-Break Galaxies as Tracers of Large-Scale Structures at High Redshifts
Sang Hyeok Im et al.
2024, ApJ, 972, 196 (NASA ADS)
Highlighted as AAS Nova¹ (see here)
2. ODIN: Rest-frame Optical Morphologies and Star Formation Activity of Ly α Emitters at $z = 2.4$, 3.1, and 4.5
Sang Hyeok Im et al.,
2026, submitted to ApJ

Second Author

1. Ongoing Morphological Transformation from Elliptical to Spiral Galaxies Induced by Gas Accretion from Neighboring Galaxies
Gene Yun, **Sang Hyeok Im** et al.,
2026, submitted to ApJ

Co-Author

1. ODIN: Searching for LyC emission from Lyman- α emitters at $z = 4.5$ in the E-COSMOS and XMM-LSS fields
Eunsuk Seo et al., (**Sang Hyeok Im**)
2025, accepted for publication in ApJ (arXiv:2603.15825)
2. ODIN: Clustering Properties of Ly α Blobs at $z \sim 2.4$ and 3.1
Byeongha Moon et al., (**Sang Hyeok Im**)
2025, Submitted to ApJ (arXiv:2601.01991)
3. ODIN: A New Lyman Alpha Blob Selection Method, Sample, and Statistical Analysis at $z \sim 3.1$
Byeongha Moon et al., (**Sang Hyeok Im**)
2025, ApJ, 998, 345 (NASA ADS)
4. ODIN: Characterizing the Three-dimensional Structure of Two Protocluster Complexes at $z=3.1$
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)

¹AAS Nova is the research highlights from the journals of the American Astronomical Society.

2025, submitted to ApJ (arXiv:2511.11826)

5. ODIN: Star Formation Histories Reveal Formative Starbursts Experienced by Ly α -emitting Galaxies at Cosmic Noon
Nicole M. Firestone et al., (**Sang Hyeok Im**)
2025, ApJL, 986, 8 (NASA ADS)
6. ODIN: High Clustering Strength of Protoclusters at Cosmic Noon
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)
2025, ApJ, 982, 74 (NASA ADS)
7. ODIN: Identifying Protoclusters and Cosmic Filaments Traced by Ly α -emitting Galaxies
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)
2024, ApJ, 977 119 (NASA ADS)
8. ODIN: Improved Narrowband Ly α Emitter Selection Techniques for $z = 2.4, 3.1$, and 4.5
Nicole M. Firestone et al., (**Sang Hyeok Im**)
2024, ApJ, 974, 217 (NASA ADS)
9. The One-hundred-deg² DECam Imaging in Narrowbands (ODIN): Survey Design and Science Goals
Kyoung-Soo Lee et al., (**Sang Hyeok Im**)
2024, ApJ, 962, 36 (NASA ADS)
10. ODIN: Where Do Ly α Blobs Live? Contextualizing Blob Environments within the Large-Scale Structure
Vandana Ramakrishnan, Byeongha Moon, **Sang Hyeok Im** et al.,
2023, ApJ, 951, 119 (NASA ADS)

GRANTS AND FELLOWSHIPS

KIAS Student Fellow (2021.09 - 2026.02)

- o Institutional Location: Korea Institute for Advanced Study, 85 Hoegi-ro, Dongdaemun-gu, Seoul 02455, Republic of Korea
- o Project Title: A 100 Deg² DECam Narrow-band Survey for the LSST Era: Tracing the Largest Cosmic Structures in the Distant Universe

PRESENTATIONS

Oral Presentations

1. **110th KAS Fall Meeting**
Date: 2024 Oct. 16 - 18
Location: Gangneung-si, Korea
Title: Testing LAEs and LBGs as Tracers of Large-Scale Structures at High Redshifts
2. **K-GMT Science Program Users Meeting 2024**
Date: 2024 Aug. 21 - 23
Location: KASI, Daejeon, Korea
Title: Ongoing Morphological Transformation from Elliptical to Spiral Galaxies Induced by Gas Accretion from Neighboring Galaxies
3. **9th Galaxy Evolution Workshop**
Date: 2023 Feb. 20 - 23
Location: Kyoto University, Japan
Title: Testing Lyman alpha emitters and Lyman-break galaxies as tracers of large-scale structures in the high-redshift universe

Poster Presentations

1. **ELT Science in Light of JWST**

Date: 2024 June 3 - 7

Location: Tohoku University, Sendai, Japan

Title: Large-scale structures in the universe traced by Lyman alpha emitters and Lyman-break galaxies

2. **108th KAS Fall Meeting**

Date: 2023 Oct. 18 - 20

Location: Ramada Plaza Jeju, Jeju, Republic of Korea

Title: Testing Lyman alpha emitters and Lyman-break galaxies as tracers of the large-scale structures in the high-z universe

3. **XXXIth General Assembly International Astronomical Union**

Date: 2022 Aug. 2 - 11

Location: BEXCO, Busan, Republic of Korea

Title: Lyman alpha emitters and Lyman-break galaxies as tracers of large-scale structures in a simulated universe

RESEARCH INTERESTS

1. Large-scale structures of the universe
2. Observational cosmology
3. Environmental effects on galaxy properties
4. Formation and evolution of proto-clusters